



Intelligent camera that can automatically detect road-side bombs



The system has been successfully demonstrated to Dutch Ministry of Defence, showing its detection capabilities on a military terrain, where it was able to detect all unknown test objects and changes with at least medium contrast (Credit: Eindhoven University of Technology)

Roadside bombs, or so-called improvised explosive devices (IEDs), are a serious threat to both troops and civilians in conflict zones. These are easily fabricated using common house-hold items; with instructions widely available online, their construction no longer requires military training.

Dennis van de Wouw and his project partners have developed a real-time early-warning system for IEDs that is mounted on a vehicle. This system can automatically detect unknown objects and suspicious changes in the environment. Automated detection of such unknown objects is very difficult using conventional detection techniques, as it is simply not known what the IEDs look like.

So, the researchers started from the assumption that placing an IED causes subtle changes to the environment, such as digging tracks or newly appeared objects, like triggers used by insurgents or terrorists to detonate the IED. Therefore van de Wouw equipped a military vehicle with an intelligent video system that can automatically find suspicious changes in the environment while driving. This system consists of a stereo camera (a camera with two lenses that generates 3D images), a GPS positioning system and an image analysis platform.

The system automatically records the environment and compares it with images recorded previously; for instance, during an earlier clearance operation. All changes are then analysed by the system. Only those changes that indicate the possible presence of an IED are presented to the military operator through an interactive graphical user interface. "This enables a military operator to take appropriate actions, such as stopping in time for further inspection of the potential threat, or to avoid it altogether," says van de Wouw.

Researchers find potential solution to overheating cellphones

Modern computer memory encodes information by switching magnetic bits within devices. Now, a ground-breaking study conducted by researchers from NUS Electrical and Computer Engineering has found a new and efficient way of using spin waves to switch magnetisation at room temperature for more energy-efficient spin memory and logic devices.

Traditional electronic chips suffer from substantial Joule heat, which occurs due to flow of an electric current producing high temperatures. It is caused by rapid motion and frequent collision among moving charges inside the devices. This not only causes a large amount of power dissipation, but also hinders the chips' processing speed and limits the number of chips that can be incorporated into appliances.

Prof. Yang Hyunsoo, team leader of the research, says, "We encounter such problems and inconveniences when using our phones, computers and other electronic devices. We often

find that these devices are becoming hot and slow; moreover, we need to charge them frequently and have to bring another portable charger sometimes."

So, rather than adopting standard electron injection methods used in traditional electronics, Prof. Yang's team creatively used spin waves to switch magnetisation. Spin waves propagate disturbances in the ordering of magnetic materials, and from the quasiparticle point of view, spin waves are known as magnons.

The team has built a bi-layer system consisting of an anti-ferromagnetic magnon transport channel and a topological insulator spin source. In a first, they then successfully demonstrated spin-wave-driven magnetisation switching in the adjacent ferromagnetic layer with a high efficiency at room temperature. The new switching scheme based on spin waves can avoid moving charges. Hence, much less Joule heat and power dissipation would be expected for devices.